K.R. Mangalam University

School of Engineering and Technology

LAB 2 BENCHMARKING REPORT

Program: BCA (AI & DS)

Course Code - ENCA351 (2026-27)

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Course: BCA (AI & DS) - Section B

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Complexity Comparison Table

4. Complexity table

Algorithm / Pattern	Theoretical time + space	Observed time (measured)	Observed memory	Growth trend
Linear Search	O(n), O(1)	6µs → 5.1ms (n=100 → 50k)	0.07–0.16KB, flat	linear
Binary Search	O(log n), O(1)	1–6µs, flat	0.02–0.18KB, flat	logarithmic
Single Loop	O(n), O(1)	5µs → 2.3ms	flat	linear
Nested Loop	O(n^2), O(1)	0.26ms → 302s	0.09KB, flat	quadratic
Factorial (Recursive)	O(n), O(n)	57µs → 2.6ms (n=100 → 900)	0.2 → 17.8KB	linear, capped below 1000
Factorial (Iterative)	O(n), O(1)	22µs → 0.26ms (n=100 → 1000)	0.2 → 2.3KB	linear

Each row read against measurement:

- Linear Search: best O(1) on a first-element hit, O(n) otherwise. Measured comparisons track n exactly (51, 501, 5001, 25001 present; n absent) and time grows 6 microseconds to 5 milliseconds. Textbook linear.
- Binary Search: best O(1) on a middle hit, O(log n) otherwise. Measured 6, 9, 13, 15 comparisons while n grows 500x, time flat at 1-6 microseconds. Logarithmic confirmed; price is sorted input.
- Single Loop: O(n) in every case. Measured 5 microseconds to 2.3 milliseconds tracking n, memory flat. Linear and lean.
- Nested Loop: O(n^2) everywhere. The ratio column against single loop runs 52x, 949x, 10034x, 129195x; at 50k it takes 302 seconds while memory stays 0.09KB. Quadratic time with constant space: time, not space, is the wall.
- Factorial (Recursive): O(n) time, O(n) stack space. At n=500 it is 8x slower and 8x more memory than iterative with identical results, and n=1000 exceeds the default recursion limit, so 900 is the max measured.
- Factorial (Iterative): O(n) time, O(1) space. Fastest everywhere measured (0.26 milliseconds at n=1000); the default choice.

Growth trends: linear rows climb with n, the binary row stays flat, and the nested row explodes - exactly the fan-out the log-scale time chart in Sec. 5 shows.